

一. 选择题 (24分)

~~1. D~~ 2. B. C. 1. D. 2. B 3. A 4. C. 5. D . 6. B.

二. 填空题 (25分)

7. $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ 8. 100 9. $\frac{1}{4}(A+2E)$ 10. 2个 11. $\begin{bmatrix} 3^n & n \cdot 3^{n-1} & 0 & 0 \\ 0 & 3^n & 0 & 0 \\ 0 & 0 & 3 \cdot 6^{n-1} & 9 \cdot 6^{n-1} \\ 0 & 0 & 6^{n-1} & 3 \cdot 6^{n-1} \end{bmatrix}$

三. 证明题

$$12. \text{ 设 } A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & & & \\ \vdots & & & \\ a_{m1} & & & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & \dots & b_{1m} \\ b_{21} & & \\ \vdots & & \\ b_{n1} & & b_{nm} \end{pmatrix}$$

分别 $\text{tr}(AB) = \underline{\hspace{2cm}}$. $\text{tr}(BA) = \underline{\hspace{2cm}}$. 因此 $\text{tr}(AB) = \text{tr}(BA)$

13. A, B 均是正定阵. 故存在可逆矩阵 P, 使 $P^T A P = I$. 而 $P^T B P$ 仍是正定阵.

对 $P^T B P$, 存在正交矩阵 R, 使

$$R^T (P^T B P) R = \text{diag}(b_1, b_2, \dots, b_n) \triangleq \Lambda \quad (b_i > 0)$$

$$\text{而} \quad R^T (P^T A P) R = R^T I R = I.$$

$$\text{令 } P R = Q, \text{ 则} \quad Q^T B Q = \Lambda, \quad Q^T A Q = I. \quad \textcircled{1}$$

$$\text{则 } Q^T (A+B) Q = Q^T A Q + Q^T B Q = I + \Lambda. \quad \textcircled{2}$$

$$\textcircled{2} \text{ 两端取行列式 } |Q^T (A+B) Q| = |Q|^2 |A+B| = |I + \Lambda| = \prod_{i=1}^n (1+b_i) \geq 1 + \prod_{i=1}^n b_i = |I| + |\Lambda|$$

$$\text{对 } \textcircled{1} \text{ 两端取行列式 } |\Lambda| = |Q|^2 |B|, \quad |I| = |Q|^2 |A|$$

$$\therefore \textcircled{2} \quad |Q|^2 |A+B| \geq |I| + |\Lambda| = (|Q|^2 (|A| + |B|)) \Rightarrow |A+B| \geq |A| + |B|.$$

四. 计算题

14. 对矩阵初等行变换:

$$A \xrightarrow{r_2-r_1, r_3-4r_1, r_4-2r_1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -3 & -2 & -3 \\ 0 & -3 & -2 & 3 \\ 0 & 3 & \mu-2 & \lambda-2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -3 & -2 & -3 \\ 0 & 0 & \mu-4 & \lambda-5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ 当 $\mu=4$ 且 $\lambda=5$ 时 $r(A)=2$.

当 $\mu \neq 4$ 或 $\lambda \neq 5$ 时, $r(A)=3$.



15. 由 $\det(A) = 1 \neq 0 \Rightarrow A$ 可逆.

$AX - 2A = XA$ 两边右乘 A^{-1} 得

$$AX - 2E = X \Rightarrow (A - E)X = 2E$$

$$\Rightarrow X = 2E(A - E)^{-1} = 2(A - E)^{-1} = 2 \begin{pmatrix} -\frac{1}{3} \begin{pmatrix} 2 & -5 \\ -1 & 1 \end{pmatrix} & 0 & 0 \\ 0 & 0 & -\frac{1}{4} \begin{pmatrix} 1 & -1 \\ -7 & 3 \end{pmatrix} \end{pmatrix} = \frac{1}{6} \begin{pmatrix} -8 & 20 & 0 & 0 \\ 4 & -4 & 0 & 0 \\ 0 & 0 & -3 & 3 \\ 0 & 0 & 21 & -9 \end{pmatrix}$$

16. (1) 按第一列展开得 $|A_n| = 2a|A_{n-1}| + a^2(-1)^{2+1}|A_{n-2}| = 2a|A_{n-1}| - a^2|A_{n-2}|$

$$\text{则 } |A_n| - a|A_{n-1}| = a(|A_{n-1}| - a|A_{n-2}|) = a^{n-2}(|A_2| - a|A_1|) = a^n$$

$$\therefore |A_n| = a^n + a|A_{n-1}| = a^n + a(a^{n-1} + a|A_{n-2}|) = 2a^n + a^2|A_{n-2}| = \dots =$$

$$(n-1)a^n + a^{n-1}|A_1| = (n+1)a^n \quad \text{（也可使用）}$$

(2) 由 Cramer 法则 $D = |A_n| \neq 0$ 时即 $a \neq 0$ 时方程组有唯一解

$$\underline{\text{且}} \quad x_1 = \frac{D_1}{D} = \frac{na^{n-1}}{(n+1)a^n} = \frac{n}{(n+1)a} \quad \cdot \quad x_n = \frac{D_n}{D} = \frac{(-1)^{n+1}(a^2)^{n-1}}{(n+1)a^n} = (-1)^{n+1} \frac{a^{n-2}}{n+1}.$$

